

Arab Academy for Science, Technology & Maritime Transport (AAST)



College of Computing & Information Technology

CSE3101: Software Requirements and Specifications

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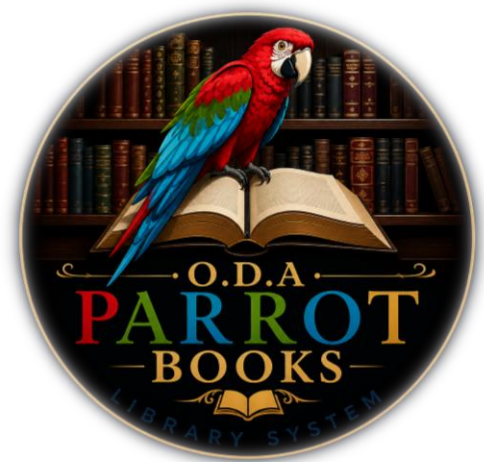
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Software Requirements Specifications Document.

PARROT BOOKS SALES SYSTEM

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Software Requirements Specifications

for

<Parrot Books Sales System>

Version 1.0 approved

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**<Arab Academy for Science, Technology, and Maritime Transport —
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Revision History

Name	Date	Reason For Changes	Version
O.D.A	29/07/2026	Initial SRS based on Parrot Books case study and Vision & Scope document	1.0

1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Release 1.0 of the Parrot Books Sales System — a web-based application that replaces Parrot Books Ltd's current manual/screen-based Sales Order Processing and Sales Ledger processes described in the “Current System Documentation” case study. This document covers the full scope of Release 1.0 as defined in the accompanying Vision and Scope Document.

1.2. Document Conventions

Functional requirements are uniquely identified as FRn (e.g., FR1) in the order they appear. Priorities stated at the feature level (High, Medium, Low) are inherited by all functional requirements within that feature.

1.3. Intended Audience and Reading Suggestions

This document is intended for the project's business analysts, developers, testers, and the course instructor/TA evaluating the project, as well as Parrot Books' sales, warehouse, and accounts management who will validate that the requirements reflect their business process. Readers new to the project should read Section 1 and Section 2 for context, then proceed to Section 3 for the detailed feature requirements most relevant to their role. Testers should pay particular attention to Section 3 (functional requirements) and Section 5 (non-functional requirements) when designing test cases.

1.4. Project Scope

The Parrot Books Sales System automates the sequence of activities described in the case study: receiving and validating a sales order, checking stock and raising purchase requisitions, preparing and dispatching the order acknowledgement, generating and actioning the picking ticket and dispatch note, and — on the Sales Ledger side — preparing and sending sales invoices and customer statements, matching incoming payments to invoices, and preparing bank deposits. The full business rationale, objectives, and release scope are defined in the Vision and Scope Document; this SRS specifies Release 1.0 as a subset of that long-term vision.

1.5. References

- Vision and Scope Document — Parrot Books Sales System, Version 1.0.
- Case Study: “Current System Documentation” — Parrot Books Ltd, Sales Order and Sales Ledger processes and sample documents (Purchase Requisition, Sales Order Acknowledgement, Picking Ticket, Dispatch Note, Purchase Order, Goods Received Note, Sales Invoice, Remittance Advice).
- 12th Week Project assignment sheet, Course CSE3101 — Software Requirements and Specifications, Arab Academy for Science, Technology, and Maritime Transport.
- Wiegers, K. & Beatty, J., Software Requirements, Third Edition, Microsoft Press — used as the methodological and terminological reference for this SRS and the Vision and Scope Document.
- Requirements Engineering Artifacts document (companion file) — requirement categorization, stakeholder analysis, context diagram, feature tree, use case diagram and scenario, state chart, dialogue map, decision tree/table, and event-response table.

2. Overall Description

2.1. Product Perspective

The Parrot Books Sales System is a new, self-contained web application that replaces the existing manual/isolated-screen Sales Order Processing and Sales Ledger systems described in the case study. It interfaces with two processes that remain manual in Release 1.0: the Purchase Order/Publisher process (the system generates requisition and purchase order data; publishers respond on paper with a Goods Received Note entered into the system) and the banking process (the system prepares the bank deposit slip data; physical deposit remains a manual cashier activity). See the Context Diagram (Level 0 DFD) and System Architecture Diagram in the companion Requirements Engineering Artifacts document.

2.2. Product Features

At a high level the product provides six feature groups, detailed fully in Section 3 and illustrated in the Feature Tree diagram (companion document):

- Sales Order Management — capture, validation, and acknowledgement of customer orders.
- Stock & Purchasing — stock-on-hand checking and purchase requisition/order generation.
- Warehouse & Fulfillment — picking ticket and dispatch note generation.
- Sales Ledger & Invoicing — invoice generation and customer statements.
- Payments & Cash Receipts — remittance matching and bank deposit preparation.
- Reporting & Administration — catalogue/customer maintenance and role-based access control.

2.3. User Classes and Characteristics

User Class	Characteristics	Frequency of Use
Sales Staff	Enter and validate customer orders; moderate technical skill; familiar with the paper order/acknowledgement layout	Daily, high volume
Warehouse & Shipping Staff	Generate and action picking tickets and dispatch notes on shared warehouse-floor terminals	Daily
Credit Control / Accounts Staff	Generate invoices and statements, match payments; needs an audit trail	Daily / monthly cycle
Cashier	Prepares bank deposit records from matched payments	Daily
Customers (Release 1.0: order placement only)	External, varying technical skill and connectivity; require a simple, guided ordering flow	Occasional, order-driven
Manager / Administrator	Views reports, manages user accounts and catalogue data; highest privilege level	Weekly

2.4. Operating Environment

The system is a browser-based web application (responsive design for desktop and tablet) deployed on a standard web/application server with a relational database backend. It must operate correctly on current versions of major browsers (Chrome, Edge, Firefox, Safari) and requires only an internet or intranet connection and no client-side installation.

2.5. Design and Implementation Constraints

- Currency and pricing must be represented and displayed in Egyptian Pounds (EGP), matching the company's existing documents.
- Screen layouts for the order, acknowledgement, picking ticket, dispatch note, invoice, and remittance advice should mirror the field layout of the existing paper documents shown in the case study, to minimize staff retraining.
- The system must group purchase requisitions/orders by publisher, consistent with current business practice ('a different requisition is raised for books from different publishers').
- Interfaces to the Purchase Order/Publisher process and to the banking process remain file/paper-compatible in Release 1.0 (see Vision and Scope, Section 1.7).
- The system must not ship partial quantities for a single order line; the firm only ships the complete quantity for an item.

2.6. User Documentation

The project will deliver a brief user guide (per role: sales, warehouse, accounts, customer) covering the core workflows, and in-application contextual help text on the order-entry and payment-matching screens. Formal training materials for staff rollout are noted as a deployment consideration in the Vision and Scope Document.

2.7. Assumptions and Dependencies

- Existing Books, Customers, and Publishers reference data can be exported from the manual system and migrated into the new database.
- Stock-on-hand (Quantity on Hand) figures are assumed to be maintained accurately in the new system once cut-over is complete; Release 1.0 does not include a separate warehouse-management/inventory-counting subsystem.
- The nominal ledger update referenced in the case study is assumed to be an existing downstream system or ledger module that receives invoice and deposit data from this system; its internal design is outside this SRS's scope.

3. System Features

Features are organized by the major services of the product, mirroring the process groupings in the case study (Sales Order System and Sales Ledger System). Each feature lists its functional requirements, uniquely numbered.

3.1. Sales Order Management

3.1.1 Description and Priority

Allows sales staff to receive a customer's sales order, validate it against the book catalogue and pricing data, and — once accepted — trigger stock checking and acknowledgement preparation. Invalid orders are referred back to the customer.

Priority: High.

3.1.2 Stimulus/Response Sequences

- Sales staff opens 'New Sales Order' and enters/imports order header and item lines → system validates each line.
- System detects an unsupplied book or incorrect pricing → system flags the line and prevents acceptance until corrected, or the order is returned to the customer.
- Order passes validation → system automatically initiates the stock-check use case (Feature 2).

3.1.3 Functional Requirements

- FR1: The system shall capture sales order header data (customer, ship-to address, customer reference, shipping terms, payment terms, delivery terms, order date) and one or more order item lines (ISBN, title, quantity, list price).
- FR2: The system shall validate that each ordered book (by ISBN) exists in the Books catalogue and is supplied by the firm, rejecting lines that reference unsupported books.
- FR3: The system shall validate that the price on each order line matches the current catalogue list price, flagging any mismatch as an error.
- FR4: The system shall allow sales staff to return an invalid order to the customer with a system-generated reason (unsupplied item and/or incorrect pricing).
- FR5: The system shall calculate the extended price for each line (quantity × price) and the order total (sum of extended prices) automatically.
- FR6: Upon successful validation, the system shall persist the sales order and its item lines and change the order status to 'Validated'.

3.2. Stock and Purchasing

3.2.1 Description and Priority

Checks Quantity on Hand for each validated order line and, where stock is insufficient, generates a purchase requisition (grouped by publisher) that feeds the manual Purchase Order process; records Goods Received Notes when stock arrives.

Priority: High.

3.2.2 Stimulus/Response Sequences

- Validated order line is checked against Quantity on Hand → system flags shortfall lines automatically.
- Shortfall detected → system pre-populates a purchase requisition with book data from the catalogue and lets sales staff complete requester/authorization fields.
- Goods Received Note is entered by warehouse staff → system updates Quantity on Hand for the received ISBN(s).

3.2.3 Functional Requirements

- FR7: The system shall display current Quantity on Hand for every item on a sales order at the time of validation.
- FR8: The system shall automatically flag any order line where the ordered quantity exceeds Quantity on Hand.
- FR9: The system shall generate a purchase requisition per publisher, listing only the shortfall items sourced from that publisher, with system-populated ISBN, title, and list price.
- FR10: The system shall allow an authorized user to convert an approved requisition into a purchase order addressed to the relevant publisher.
- FR11: The system shall allow recording of a Goods Received Note (publisher, date received, carrier, items and quantities) and shall increase Quantity on Hand accordingly upon confirmation.
- FR12: The system shall not allow an order to proceed to acknowledgement while any of its lines remain in shortfall, unless overridden by an authorized user with a recorded reason.

3.3. Order Acknowledgement, Picking and Dispatch

3.3.1 Description and Priority

Prepares and sends the sales order acknowledgement once stock is confirmed available, then supports the warehouse in generating a picking ticket and the shipping department in generating a dispatch note from the same order record, eliminating the manual re-keying described in the case study.

Priority: High.

3.3.2 Stimulus/Response Sequences

- *Order confirmed with sufficient stock → system generates the acknowledgement (header + item detail) automatically from the validated order.*
- *Acknowledgement generated → system notifies the warehouse and auto-populates a picking ticket listing only in-stock, available items.*
- *Warehouse confirms items picked → system enables generation of the dispatch note pre-filled from the picking ticket.*

3.3.3 Functional Requirements

- FR13: The system shall generate a sales order acknowledgement containing a header section (customer, ship-to, shipping/payment/delivery terms, sales order number, order date) and an item-detail section (ISBN, title, quantity, price each, extended price, order total).
- FR14: The system shall distribute the acknowledgement to the customer, the warehouse, and the credit control function, and shall retain a copy against the sales order record.
- FR15: The system shall generate a picking ticket listing only items available in the warehouse for that order and shall not include items still on backorder.
- FR16: The system shall not allow a dispatch note to be created for quantities greater than those confirmed on the picking ticket.
- FR17: The system shall generate a dispatch note pre-populated from the picking ticket and additional shipment data (carrier, dispatch date), and shall distribute copies to the customer, the sales ledger function, and the shipping department's own records.

3.4. Sales Ledger and Invoicing

3.4.1 Description and Priority

Triggered automatically by dispatch note creation, generates the sales invoice from existing order, publisher, and book data, updates the sales/nominal ledger, and produces monthly customer statements.

Priority: High.

3.4.2 Stimulus/Response Sequences

- *Dispatch note recorded → system automatically drafts a sales invoice using data already captured on the order and dispatch note.*
- *Invoice approved → system updates the sales ledger and (via interface) the nominal ledger, and sends the invoice to the customer.*
- *Monthly statement cycle runs → system compiles all unpaid invoices and current balance per customer into a statement.*

3.4.3 Functional Requirements

- FR18: The system shall generate a sales invoice automatically upon dispatch note creation, populated with dispatch note, order, and book/price data, requiring no re-keying of item details.
- FR19: The system shall record each generated invoice against the sales ledger and provide the data needed to update the nominal ledger.
- FR20: The system shall distribute the invoice to the customer and retain a copy on the invoice record.
- FR21: The system shall, on a monthly cycle, generate a customer statement listing all outstanding invoices and the total balance due for every customer with activity or an outstanding balance in that period.
- FR22: The system shall distribute each statement to the relevant customer and retain a copy on file.

3.5. Payments and Cash Receipts

3.5.1 Description and Priority

Supports matching an incoming cheque and remittance advice against outstanding invoices, marking invoices as paid, and preparing the cashier's bank deposit slip.

Priority: Medium.

3.5.2 Stimulus/Response Sequences

- *Cheque and remittance advice received → system displays the customer's outstanding invoices for matching against the remittance detail.*
- *Match confirmed → system marks the referenced invoices as paid and reduces the customer's outstanding balance.*
- *End of day → cashier requests a deposit slip → system compiles all confirmed cheques received that day into a single deposit record.*

3.5.3 Functional Requirements

- FR23: The system shall allow accounts staff to record a received cheque and remittance advice against a customer account.
- FR24: The system shall display outstanding invoices for that customer to support matching against the remittance advice's listed invoices and amounts.
- FR25: The system shall flag a discrepancy where the cheque amount and the sum of remittance-advice invoice amounts do not match, preventing automatic invoice closure until resolved.
- FR26: Upon confirmed match, the system shall mark each referenced invoice as paid and update the customer's outstanding balance and the sales ledger.
- FR27: The system shall compile all same-day confirmed payments into a bank deposit record for the cashier and shall provide the data needed to update the nominal ledger's bank and debtors balances.

3.6. Reporting and Administration

3.6.1 Description and Priority

Provides role-based access control and basic operational reporting/catalogue-and-customer maintenance to support the other features.

Priority: Medium.

3.6.2 Stimulus/Response Sequences

- *Administrator logs in → system presents catalogue, customer, publisher, and user-management screens.*
- *Manager requests a report → system produces order-status, stock, and receivables summaries.*

3.6.3 Functional Requirements

- FR28: The system shall enforce role-based access control so that each user class can access only the screens and actions appropriate to its role.
- FR29: The system shall allow an administrator to maintain (add/update/deactivate) Books, Customers, and Publishers reference data.
- FR30: The system shall provide summary reports covering open orders by status, current stock levels, and outstanding customer balances.
- FR31: The system shall log key actions (order validation, invoice generation, payment recording) with user and timestamp for audit purposes.

4. External Interface Requirements

4.1. User Interfaces

The system provides role-specific web screens: Sales Order Entry, Stock/Requisition, Picking Ticket, Dispatch Note, Sales Invoice, Customer Statement, Payment Matching, and an Administration/Reports dashboard. Screen field layouts mirror the equivalent paper documents in the case study (e.g., the Sales Order Acknowledgement screen mirrors the header/item-detail structure of the sample document) so that staff already familiar with the paper forms can transition with minimal training. All screens follow a common layout with a persistent navigation sidebar, a page header showing the record's status, and standard action buttons (Save, Validate, Print/Export, Cancel). See the Dialogue Map in the companion Requirements Engineering Artifacts document for the full screen-navigation flow.

4.2. Hardware Interfaces

No special-purpose hardware is required for Release 1.0. The system is accessed via standard workstations, laptops, or tablets with a modern web browser; no barcode scanners or warehouse-automation hardware are assumed, though standard office printers are used to print acknowledgements, picking tickets, and dispatch notes where a physical copy is still required.

4.3. Software Interfaces

Interface	Direction	Purpose
Relational database (e.g., PostgreSQL/MySQL)	Internal	Persistent storage of orders, books, customers, publishers, invoices, and payments.
Nominal Ledger / accounting system	Outbound (data export)	Receives invoice and bank-deposit data to update the general ledger.
Email/PDF generation service	Outbound	Delivers acknowledgements, invoices, and statements to customers.
Purchase Order / Publisher process	Outbound (Release 1.0: file/print)	Sends purchase requisition and purchase-order data; Goods Received Notes are entered back manually.

4.4. Communications Interfaces

All client-server communication uses HTTPS (TLS-encrypted HTTP) over the company's intranet or the public internet. Outbound customer communications (acknowledgements, invoices, statements) may be sent by email using standard SMTP with attachments in PDF format. No other communication protocols are required for Release 1.0.

5. Other Nonfunctional Requirements

5.1. Performance Requirements

- NFR-PERF1: Common screens (order entry, stock check, invoice generation) shall respond to user actions within 2 seconds under normal load.
- NFR-PERF2: The system shall support at least 50 concurrent staff users and a reasonable volume of concurrent customer order submissions without degradation of the 2-second response target.
- NFR-PERF3: Monthly statement generation for the full customer base shall complete within a defined overnight batch window without impacting daytime transactional performance.

5.2. Safety Requirements

- NFR-SAFE1: Not applicable in the sense of physical safety, as this is a business information system.
- NFR-SAFE2: The system shall prevent an order from being dispatched or invoiced twice, and shall prevent stock and ledger updates from being applied more than once for the same event (idempotent processing).

5.3. Security Requirements

- NFR-SEC1: The system shall require authenticated login for all staff and administrative functions; customers placing orders shall authenticate or verify identity by order/customer reference.
- NFR-SEC2: The system shall enforce role-based authorization so that, for example, only credit-control staff can mark invoices as paid.
- NFR-SEC3: All data in transit shall be encrypted (HTTPS/TLS); sensitive fields shall be encrypted at rest.
- NFR-SEC4: The system shall maintain an audit log of who created, modified, or approved each order, invoice, and payment record.

5.4. Software Quality Attributes

- QA1: Usability: staff should be able to complete a routine order-to-acknowledgement task with no more than brief, role-specific training, given the deliberate similarity to existing paper forms.
- QA2: Reliability: the system shall be available at least 99.5% of scheduled operating hours, excluding planned maintenance windows.
- QA3: Maintainability: the codebase shall follow a layered architecture (presentation, business logic, data access) to allow features to be modified independently.
- QA4: Scalability: the architecture shall allow horizontal scaling of the web/application tier as order volume grows internationally.
- QA5: Portability: the browser-based client shall run unmodified on Windows, macOS, and common Linux distributions on any of the supported browsers.

6. Other Requirements

- Database requirements: the schema must represent Books, Customers, Publishers, Sales Orders/Order Items, Purchase Requisitions/Orders, Picking Tickets, Dispatch Notes, Invoices, and Payments as distinct, related entities (see the Database Design in the companion Requirements Engineering Artifacts document).
- Internationalization: all monetary values are represented in EGP for Release 1.0; the data model should not preclude adding multi-currency support in a later release (see Vision and Scope, Section 2.3).
- Legal/compliance: customer and payment-related data shall be handled in line with standard data-protection practice (access controls, audit logging) as described in Section 5.3.

Appendix A: Glossary

Term	Definition
Sales Order	A customer's request to purchase one or more books, subject to validation before acceptance.
Quantity on Hand	The quantity of a given book is currently available in warehouse stock.
Purchase Requisition	An internal request, grouped by publisher, to replenish stock for items short on an accepted order.
Picking Ticket	A warehouse document listing the items and quantities to be physically retrieved for an order.
Dispatch Note	A shipping document confirming the items sent to the customer for a given order.
Sales Invoice	A billing document raised once goods are dispatched, requesting customer payment.
Remittance Advice	A document accompanying customer payment identifying which invoices are being paid.
Nominal Ledger	The company's general accounting ledger, updated with invoice and deposit data.

Appendix B: Analysis Models

The full set of analysis and design models — Level 0 Context Diagram, Feature Tree, Use Case Diagram and Usage Scenario, State Chart Diagram and State Table, Dialogue Map, Decision Tree, Decision Table, and Event–Response Table — are provided in the companion document “Requirements Engineering Artifacts — Parrot Books Sales System”.>

Appendix C: Issues List

- TBD: Final list of supported browsers/minimum versions to be confirmed with the course instructor's grading environment.
- TBD: Whether Release 1.0 must support any currency besides EGP for international customers, or whether this is deferred entirely to Release 3.0 as currently assumed.
- TBD: Exact nominal ledger interface format (file export vs. direct database link), pending confirmation of the existing ledger system referenced in the case study.